

(ICMPv4)

Internet Control Message Protocol Version 4

Outline

Introduction

Messages

Debugging Tools

INTRODUCTION

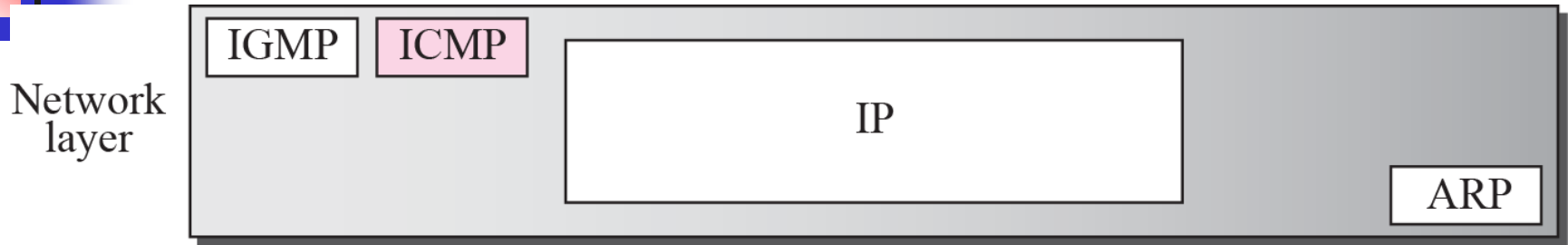
The IP protocol has no error-reporting or error correcting mechanism.(unreliable protocol)

- ❑ What happens if something goes wrong?
- ❑ What happens if a router must discard a datagram
 - because it cannot find router to the final destination
 - because the time-to-live field has a zero value

IP also lacks a mechanism for host and management queries-to determine if router or another host is alive etc.

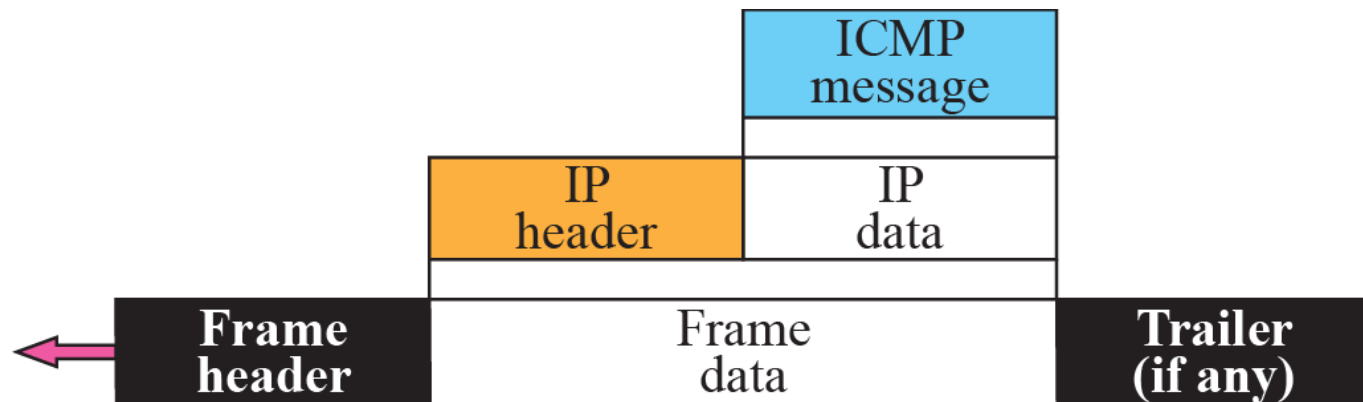
IP protocol in itself has no built-in mechanism to handle such situation notify the original host.

Position of ICMP in the network layer



ICMP in itself is network layer protocol, its messages are not directly passed to the data link layer but will be encapsulated inside datagram (value of protocol field in IP datagram is 1 to indicate IP data is an ICMP message)

ICMP encapsulation



ICMP messages are divided into two categories:

- ✓ **error-reporting messages** and
- ✓ **query messages.**

error-reporting messages

report problems that a router or a host (destination) may encounter when it processes an IP packet.

query messages,

- help a host or a network manager get specific information from a router or another host.
- hosts can discover and learn about routers on their network
- routers can help a node redirect its messages.

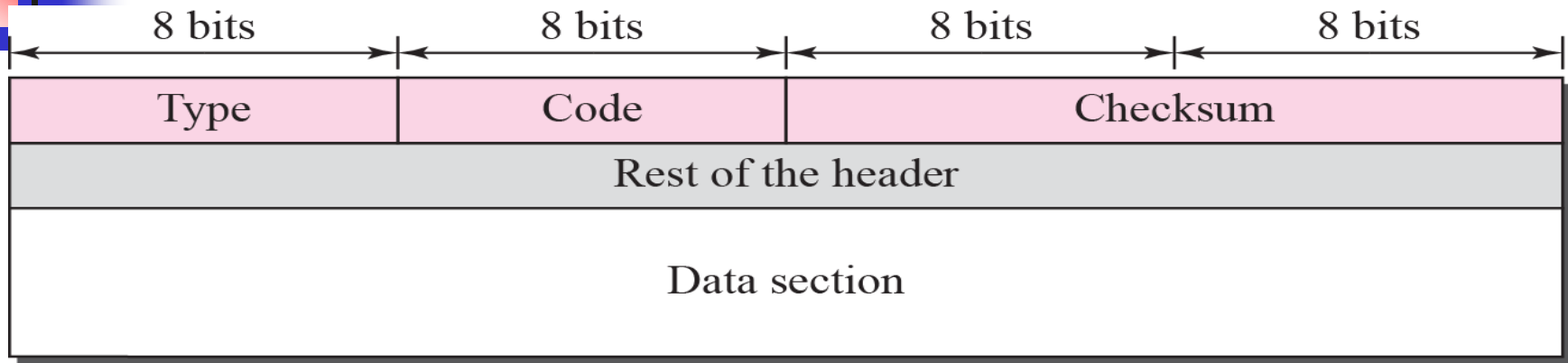
Table 9.1 *ICMP messages*

<i>Category</i>	<i>Type</i>	<i>Message</i>
Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply

Type	Code	Meaning
0	0	Echo Reply
3	0	Net Unreachable
	1	Host Unreachable
	2	Protocol Unreachable
	3	Port Unreachable
	4	Frag needed and DF set
	5	Source route failed
	6	Dest network unknown
	7	Dest host unknown
	8	Source host isolated
	9	Network admin prohibited
	10	Host admin prohibited
	11	Network unreachable for TOS
	12	Host unreachable for TOS
	13	Communication admin prohibited
4	0	Source Quench (Slow down/Shut up)

Type	Code	Meaning
5	0	Redirect datagram for the network
	1	Redirect datagram for the host
	2	Redirect datagram for the TOS & Network
	3	Redirect datagram for the TOS & Host
8	0	Echo
9	0	Router advertisement
10	0	Router selection
11	0	Time To Live exceeded in transit
	1	Fragment reassemble time exceeded
12	0	Pointer indicates the error (Parameter Problem)
	1	Missing a required option (Parameter Problem)
	2	Bad length (Parameter Problem)
13	0	Time Stamp
14	0	Time Stamp Reply
15	0	Information Request
16	0	Informaiton Reply
17	0	Address Mask Request
18	0	Address Mask Reply
30	0	Traceroute (Tracert)

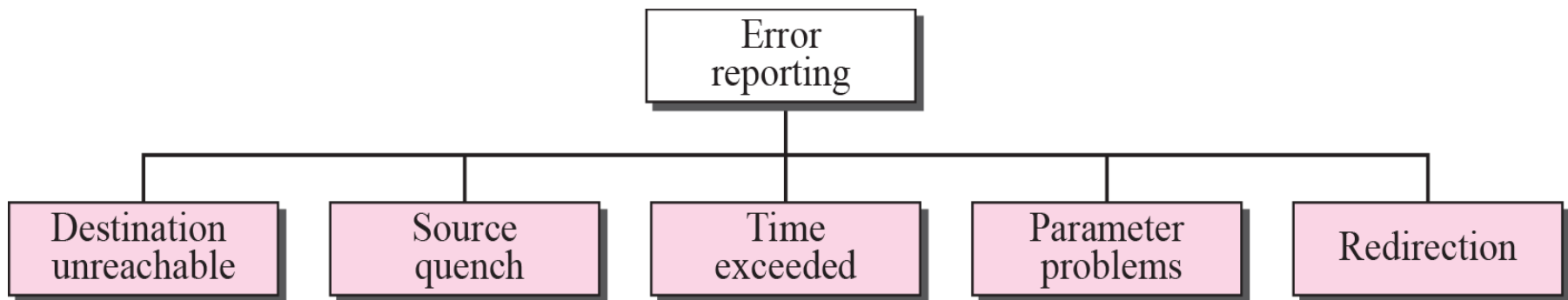
General format of ICMP messages



Though the general format of the header is different for each message type, the **first four bytes are common to all**

ICMP always reports error messages to the original source.

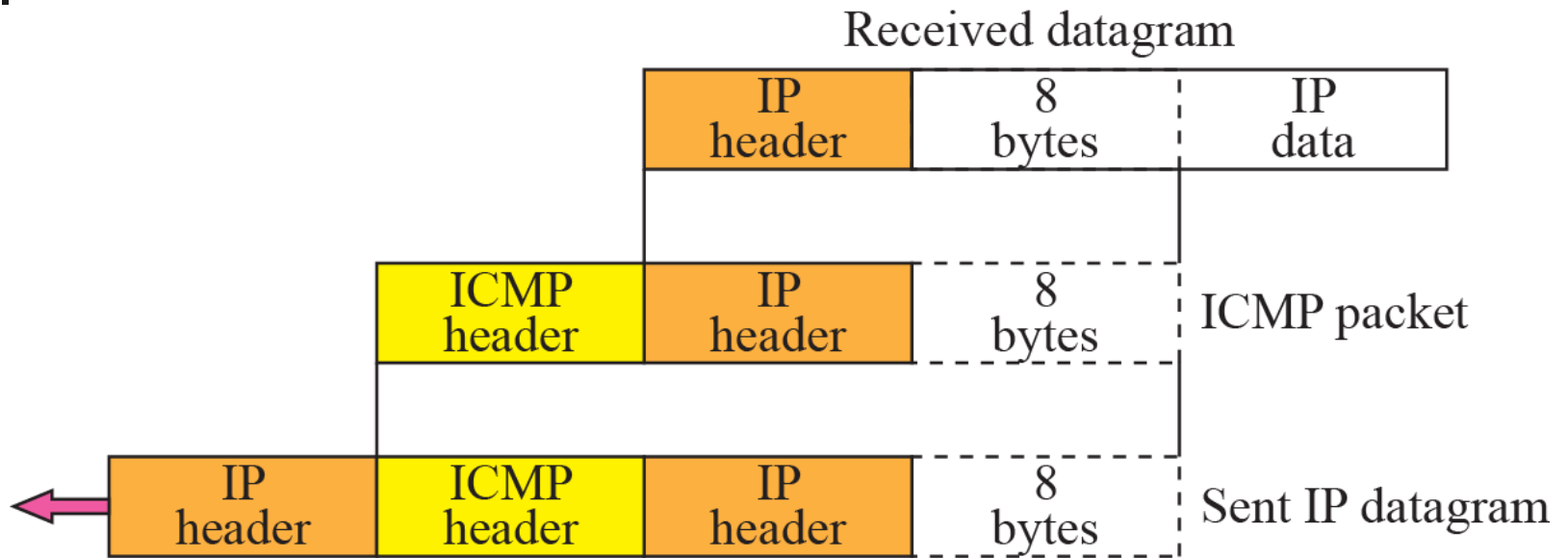
Error-reporting messages



The following are important points about ICMP error messages:

- ❑ No ICMP error message will be generated in response to a datagram carrying an ICMP error message.
- ❑ No ICMP error message will be generated for a fragmented datagram that is not the first fragment.
- ❑ No ICMP error message will be generated for a datagram having a multicast address.
- ❑ No ICMP error message will be generated for a datagram having a special address such as 127.0.0.0 or 0.0.0.0.

Contents of data field for the error message



Destination-unreachable format

Type: 3	Code: 0 to 15	Checksum
Unused (All 0s)		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

***Destination-unreachable messages with codes 2 or 3
can be created only by the destination host.***

***Other destination-unreachable messages can be
created only by routers.***

***A router cannot detect all problems that prevent the
delivery of a packet.***

- ❑ **Code 0.** The network is unreachable, possibly due to hardware failure.
- ❑ **Code 1.** The host is unreachable. This can also be due to hardware failure.
- ❑ **Code 2.** The protocol is unreachable. An IP datagram can carry data belonging to higher-level protocols such as UDP, TCP, and OSPF. If the destination host receives a datagram that must be delivered, for example, to the TCP protocol, but the TCP protocol is not running at the moment, a code 2 message is sent.
- ❑ **Code 3.** The port is unreachable. The application program (process) that the datagram is destined for is not running at the moment.
- ❑ **Code 4.** Fragmentation is required, but the DF (do not fragment) field of the datagram has been set. In other words, the sender of the datagram has specified that the datagram not be fragmented, but routing is impossible without fragmentation.
- ❑ **Code 5.** Source routing cannot be accomplished. In other words, one or more routers defined in the source routing option cannot be visited.

- ❑ **Code 6.** The destination network is unknown. This is different from code 0. In code 0, the router knows that the destination network exists, but it is unreachable at the moment. For code 6, the router has no information about the destination network.
- ❑ **Code 7.** The destination host is unknown. This is different from code 1. In code 1, the router knows that the destination host exists, but it is unreachable at the moment. For code 7, the router is unaware of the existence of the destination host.
- ❑ **Code 8.** The source host is isolated.
- ❑ **Code 9.** Communication with the destination network is administratively prohibited.
- ❑ **Code 10.** Communication with the destination host is administratively prohibited.
- ❑ **Code 11.** The network is unreachable for the specified type of service. This is different from code 0. Here the router can route the datagram if the source had requested an available type of service.

- ❑ **Code 12.** The host is unreachable for the specified type of service. This is different from code 1. Here the router can route the datagram if the source had requested an available type of service.
- ❑ **Code 13.** The host is unreachable because the administrator has put a filter on it.
- ❑ **Code 14.** The host is unreachable because the host precedence is violated. The message is sent by a router to indicate that the requested precedence is not permitted for the destination.
- ❑ **Code 15.** The host is unreachable because its precedence was cut off. This message is generated when the network operators have imposed a minimum level of precedence for the operation of the network, but the datagram was sent with a precedence below this level.

Source Quench

There is no flow-control or congestion-control mechanism in the IP protocol.

Source Quench is a kind of flow control and congestion control to the IP.

Serves two purposes

- Inform source that datagram has been discarded
- Inform source that there is a congestion in the network

Source-quench format

Type: 4	Code: 0	Checksum
Unused (All 0s)		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

One source-quench message is sent for each datagram that is discarded due to congestion.



Time-exceeded

When Datagram is discarded, a time exceeded message must be sent by router to a sender

Generated in two cases

- 1. Because of Routers**
- 2. Because not all fragments reaching in certain time limit**

Whenever a router decrements a datagram with a time-to-live value to zero, it discards the datagram and sends a time-exceeded message to the original source.

When the final destination does not receive all of the fragments in a set time, it discards the received fragments and sends a time-exceeded message to the original source.

Time-exceeded message format

Type: 11	Code: 0 or 1	Checksum
Unused (All 0s)		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

In a time-exceeded message, code 0 is used only by routers to show that the value of the time-to-live field is zero.

Code 1 is used only by the destination host to show that not all of the fragments have arrived within a set time.

Parameter-problem

Any Ambiguity in the header part of datagram create a problem

A parameter-problem message can be created by a router or the destination host.

Parameter-problem message format

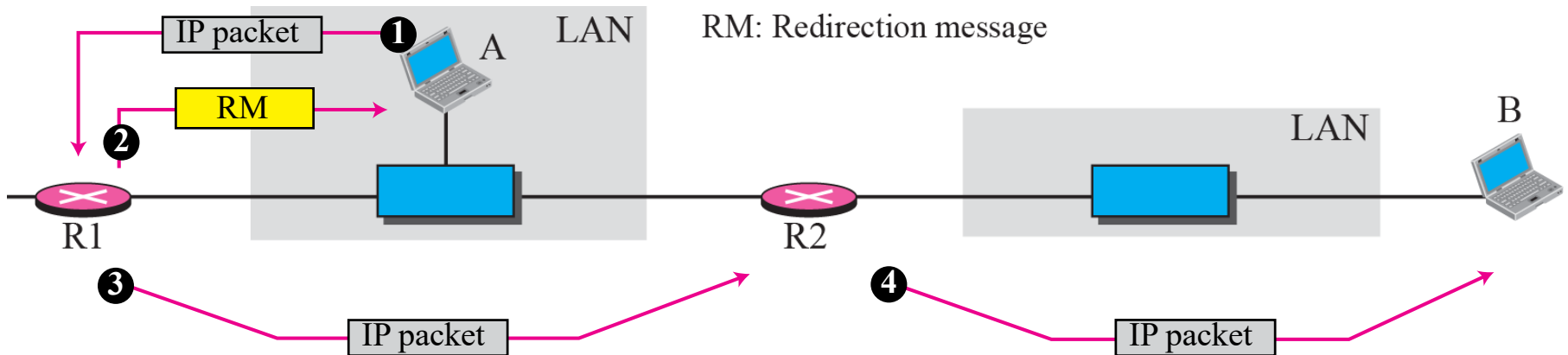
Type: 12	Code: 0 or 1	Checksum
Pointer	Unused (All 0s)	
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

- ❑ **Code 0.** There is an error or ambiguity in one of the header fields. In this case, the value in the pointer field points to the byte with the problem. For example, if the value is zero, then the first byte is not a valid field.
- ❑ **Code 1.** The required part of an option is missing. In this case, the pointer is not used.

Redirection concept

A host usually starts with a small routing table that is gradually augmented and updated.

One of the tools to accomplish this is the redirection message.



A redirection message is sent from a router to a host on the same local network.

Redirection message format

Type: 5	Code: 0 to 3	Checksum
IP address of the target router		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

- ❑ **Code 0.** Redirection for a network-specific route.
- ❑ **Code 1.** Redirection for a host-specific route.
- ❑ **Code 2.** Redirection for a network-specific route based on a specified type of service.
- ❑ **Code 3.** Redirection for a host-specific route based on a specified type of service.

Query Messages

Through query messages, ICMP diagnose some network problems

Two pairs are used today

1. Echo request and reply

2. Timestamp request and reply

An echo-request message can be sent by a host or router.

An echo-reply message is sent by the host or router that receives an echo-request message.

Echo-request and echo-reply messages can be used by network managers to check the operation of the IP protocol.

Echo-request and echo-reply messages can test the reachability of a host.

This is usually done by invoking the ping command.

Timestamp-request and timestamp-reply

Timestamp-request and timestamp-reply messages can be used to calculate the round-trip time between a source and a destination machine

Figure 9.13 *Timestamp-request and timestamp-reply message format*

Type 13: request

Type 14: reply

Type: 13 or 14	Code: 0	Checksum
Identifier		Sequence number
Original timestamp		
Receive timestamp		
Transmit timestamp		

Round Trip Time

Sending time = receive timestamp – original timestamp

Receiving time = returned timestamp – transit timestamp

Round Trip Time = sending time + receiving time

Example

Original timestamp : 46
Transit timestamp : 60

receive timestamp: 59
return time : 67

Round trip time as follows

Sending time = receive timestamp – original timestamp

Receiving time = returned timestamp – transit timestamp

Round Trip Time = sending time + receiving time

Round trip time as follows

Sending time = $59 - 46 = 13$ milliseconds

Receiving time = $67 - 60$ milliseconds

Round Trip Time = $13 + 7 = 20$ milliseconds

Time difference = receive timestamp – (original timestamp + one way time)

Time difference = $59 - (46 + 10) = 3$ milliseconds

Checksum

checksum calculation for a simple echo-request message.

We randomly chose the identifier to be 1 and the sequence number to be 9. The message is divided into 16-bit (2-byte) words.

Type 8: Echo request

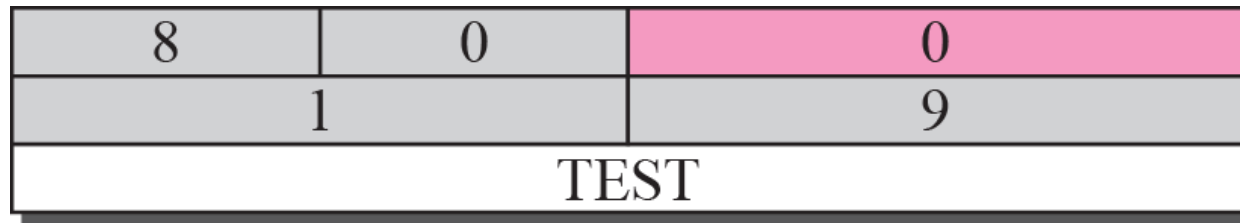
Type 0: Echo reply

Type: 8 or 0	Code: 0	Checksum
Identifier		Sequence number
Optional data Sent by the request message; repeated by the reply message		

8	0	0
1		9
TEST		

8 & 0	→	00001000	00000000
0	→	00000000	00000000
1	→	00000000	00000001
9	→	00000000	00001001
T & E	→	01010100	01000101
S & T	→	01010011	01010100
Sum	→	10101111	10100011
Checksum	→	01010000	01011100

Figure 9.14 *Example of checksum calculation*



8 & 0	→	00001000	00000000
0	→	00000000	00000000
1	→	00000000	00000001
9	→	00000000	00001001
T & E	→	01010100	01000101
S & T	→	01010011	01010100
Sum	→	10101111	10100011
Checksum	→	01010000	01011100

Type 8: Echo request

Type 0: Echo reply

Type: 8 or 0	Code: 0	Checksum
Identifier		Sequence number
Optional data Sent by the request message; repeated by the reply message		

DEBUGGING TOOLS

two tools that use ICMP for debugging:

**ping and
traceroute.**

Use the ping program to test the server fhda.edu.

The result is shown below:

```
$ ping fhda.edu
PING fhda.edu (153.18.8.1) 56 (84) bytes of data.
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=0    ttl=62    time=1.91 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=1    ttl=62    time=2.04 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=2    ttl=62    time=1.90 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=3    ttl=62    time=1.97 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=4    ttl=62    time=1.93 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=5    ttl=62    time=2.00 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=6    ttl=62    time=1.94 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=7    ttl=62    time=1.94 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=8    ttl=62    time=1.97 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=9    ttl=62    time=1.89 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=10   ttl=62    time=1.98 ms

--- fhda.edu ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10103 ms
rtt min/avg/max = 1.899/1.955/2.041 ms
```

Example

Check if the adelphia.net mail server is alive and running.

The result is shown below:

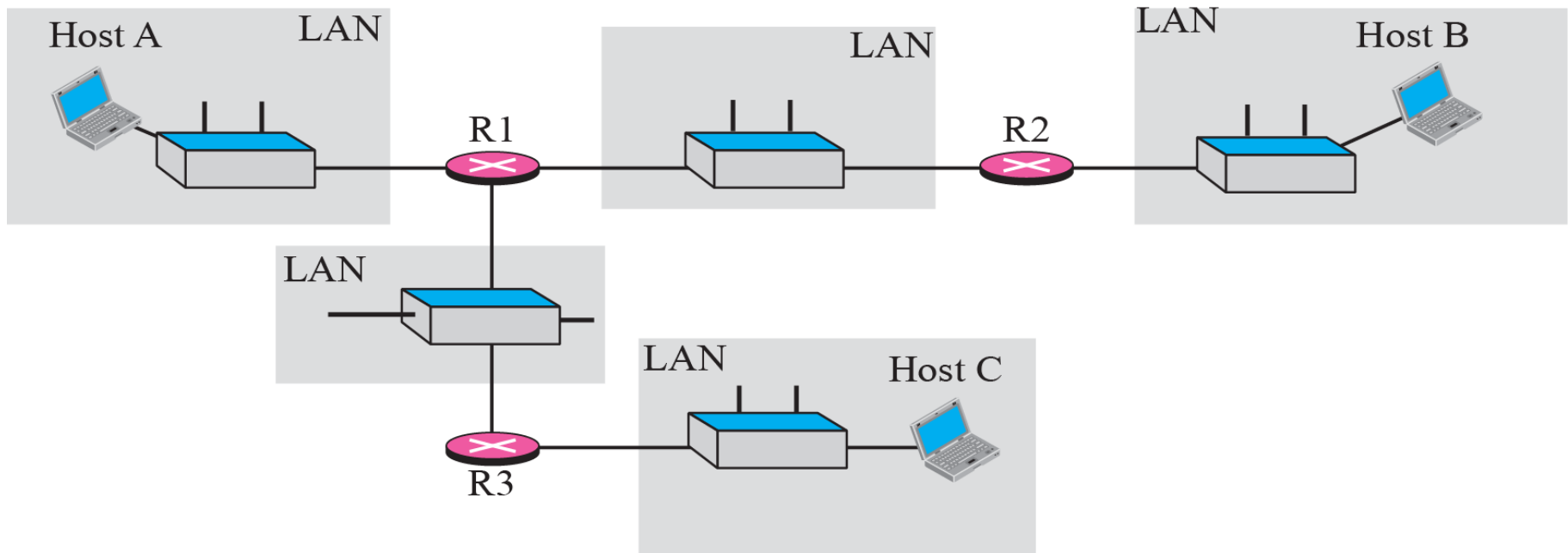
We sent 14 packets, but only 13 have been returned.

We may have interrupted the program before the last packet, with sequence number 13, was returned.

```
$ ping mail.adelphia.net
PING mail.adelphia.net (68.168.78.100) 56(84) bytes of data.
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=0    ttl=48    time=85.4 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=1    ttl=48    time=84.6 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=2    ttl=48    time=84.9 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=3    ttl=48    time=84.3 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=4    ttl=48    time=84.5 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=5    ttl=48    time=84.7 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=6    ttl=48    time=84.6 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=7    ttl=48    time=84.7 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=8    ttl=48    time=84.4 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=9    ttl=48    time=84.2 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=10   ttl=48    time=84.9 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=11   ttl=48    time=84.6 ms
64 bytes from mail.adelphia.net (68.168.78.100): icmp_seq=12   ttl=48    time=84.5 ms

--- mail.adelphia.net ping statistics ---
14 packets transmitted, 13 received, 7% packet loss, time 13129 ms
rtt min/avg/max/mdev = 84.207/84.694/85.469
```

Figure 9.15 *The traceroute program operation*



Use the **traceroute** program to find the route from the computer **voyager.deanza.edu** to the server **fhda.edu**.

The following shows the result.

```
$ traceroute fhda.edu
```

```
traceroute to fhda.edu (153.18.8.1), 30 hops max, 38 byte packets
```

1 Dcore.fhda.edu	(153.18.31.25)	0.995 ms	0.899 ms	0.878 ms
2 Dbackup.fhda.edu	(153.18.251.4)	1.039 ms	1.064 ms	1.083 ms
3 tiptoe.fhda.edu	(153.18.8.1)	1.797 ms	1.642 ms	1.757 ms

**This example we trace a longer route,
the route to xerox.com.**
The following is a partial listing.

```
$ traceroute xerox.com
```

```
traceroute to xerox.com (13.1.64.93), 30 hops max, 38 byte packets
```

1	Dcore.fhda.edu	(153.18.31.254)	0.622 ms	0.891 ms	0.875 ms
2	Ddmz.fhda.edu	(153.18.251.40)	2.132 ms	2.266 ms	2.094 ms
3	Cinic.fhda.edu	(153.18.253.126)	2.110 ms	2.145 ms	1.763 ms
4	cenic.net	(137.164.32.140)	3.069 ms	2.875 ms	2.930 ms
5	cenic.net	(137.164.22.31)	4.205 ms	4.870 ms	4.197 ms
6	cenic.net	(137.164.22.167)	4.250 ms	4.159 ms	4.078 ms
7	cogentco.com	(38.112.6.225)	5.062 ms	4.825 ms	5.020 ms
8	cogentco.com	(66.28.4.69)	6.070 ms	6.207 ms	5.653 ms
9	cogentco.com	(66.28.4.94)	6.070 ms	5.928 ms	5.499 ms

An interesting point is

- ✓ a host can send a traceroute packet to itself.
- ✓ This can be done by specifying the host as the destination.
- ✓ The packet goes to the loopback address as we expect.

```
$ traceroute voyager.deanza.edu
```

```
traceroute to voyager.deanza.edu (127.0.0.1), 30 hops max, 38 byte packets
```

1 voyager	(127.0.0.1)	0.178 ms	0.086 ms	0.055 ms
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Use the traceroute program to find the route between fhda.edu and mhhe.com (McGraw-Hill server).

Notice - we cannot find the whole route.

When traceroute does not receive a response within 5 seconds, it prints an asterisk to signify a problem (not in this case), and then tries the next hop.

```
$ traceroute mhhe.com
traceroute to mhhe.com (198.45.24.104), 30 hops max, 38 byte packets
 1  Dcore.fhda.edu      (153.18.31.254)      1.025 ms    0.892 ms    0.880 ms
 2  Ddmz.fhda.edu      (153.18.251.40)      2.141 ms    2.159 ms    2.103 ms
 3  Cinic.fhda.edu     (153.18.253.126)     2.159 ms    2.050 ms    1.992 ms
 4  cenic.net          (137.164.32.140)     3.220 ms    2.929 ms    2.943 ms
 5  cenic.net          (137.164.22.59)      3.217 ms    2.998 ms    2.755 ms
 6  SanJose1.net       (209.247.159.109)    10.653 ms   10.639 ms   10.618 ms
 7  SanJose2.net       (64.159.2.1)         10.804 ms   10.798 ms   10.634 ms
 8  Denver1.Level3.net (64.159.1.114)       43.404 ms   43.367 ms   43.414 ms
 9  Denver2.Level3.net (4.68.112.162)       43.533 ms   43.290 ms   43.347 ms
10  unknown            (64.156.40.134)      55.509 ms   55.462 ms   55.647 ms
11  mcleodusa1.net     (64.198.100.2)        60.961 ms   55.681 ms   55.461 ms
12  mcleodusa2.net     (64.198.101.202)     55.692 ms   55.617 ms   55.505 ms
13  mcleodusa3.net     (64.198.101.142)     56.059 ms   55.623 ms   56.333 ms
14  mcleodusa4.net     (209.253.101.178)    297.199 ms  192.790 ms  250.594 ms
15  eppg.com           (198.45.24.246)      71.213 ms   70.536 ms   70.663 ms
16  ...                ...                  ...         ...         ...
```

Lecture videos on the topic can be found at

ICMP 1 Introduction

<https://youtu.be/BJ7klOSeTDM>

ICMP 2 Error Reporting

https://youtu.be/HQIeY_nnlNI

ICMP 3 Timestamp Query calculations

<https://youtu.be/qgComS1NO84>

ICMP-4 Debugging tools

<https://youtu.be/rfJJITCqyOs>